Topological Memory.

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Introduction

Today:

- Proof.
  1. Walktrough the math
  2. A
  3. B
- New experimental results.
- Answering the code teleportation.

Our Code

Code

In this session, we consider LDPC codes, where each bit adjoins to Δ_1 checks, and each check adjoins to Δ_2 bits. We denote by $\Delta = \Delta_1 \Delta_2$.

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1-p)$-depolarized channel:

- There exists function $M : (0,1) \to (0,1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p$.
- There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n,k)-Toric, then both the swift rule-and the flip upon majority are local decoders.

Storing Logical State

Logical State

For a random variable X we denote by:

1. $\mathbf{E}_{\sim p}[X]$ - its expectation when measured after a single application of the p-depolarizing channel.
2. $\mathbf{E}_{\sim q^{(t)}}[X]$ - its expectation when measured after t rounds, each initiated by a p-depolarized channel followed by an application of local decoder D.

The Theorem

Theorem

Let f be any function with bounded value, denote by $|f|$ it's supremum, Then:

$$\mathbf{E}_{\sim q^{(t)}}[f] \leq \mathbf{E}_{\sim p}[f] + t|f|e^{-\Theta(\sqrt{n})}$$

Proof

By induction on t.

$$\begin{aligned}
\mathbf{E}_{\sim q^{(t)}}[f] &= \mathbf{E}_{\sim q^{(t-1)}}[\mathbf{E}[f \circ D] \sim p] \\
&\leq \mathbf{E}_{\sim p}[\mathbf{E}[f \circ D] \sim p] + (t-1)|\mathbf{E}_p[f \circ D]|e^{-c\sqrt{n}} \\
&\leq \mathbf{E}_{\sim p}[f] + t|f|e^{-\Theta(\sqrt{n})}
\end{aligned}$$